# Costs Terms, Concepts and Classifications

Chapter Two

- Why should we learn costs?
  - An important part of information system
  - A strategy
  - An example-----Airline company

- Different costs for different purposes

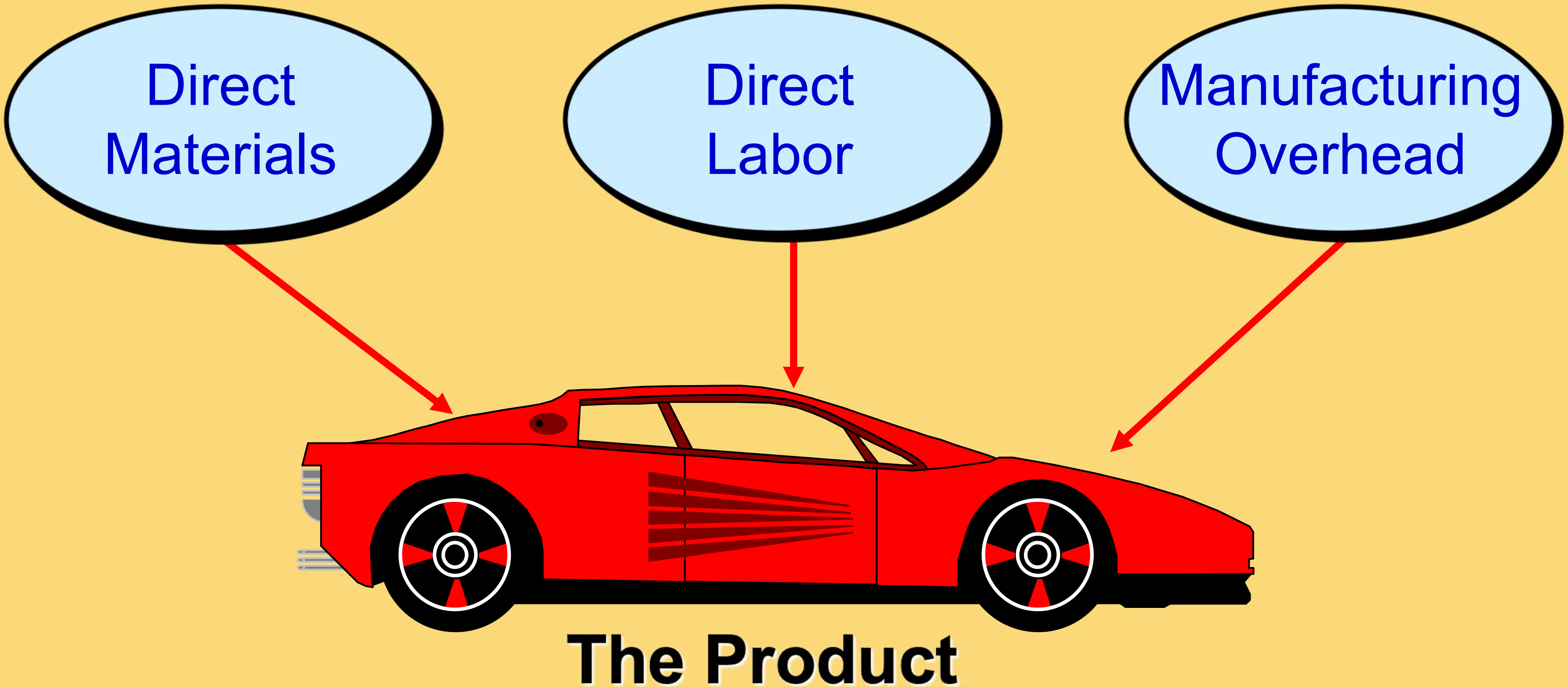
  ------costs used in financial and managerial accounting

  - Preparing financial reports
  - Assigning costs to cost objects
  - Predicting cost behavior
  - Making business decisions

# Learning Objective 1

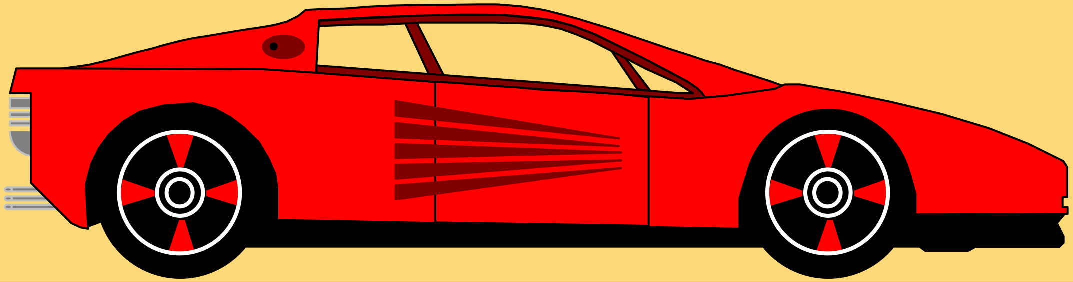
Identify and give examples of each of the three basic manufacturing cost categories.

# Manufacturing Costs



# Direct Materials

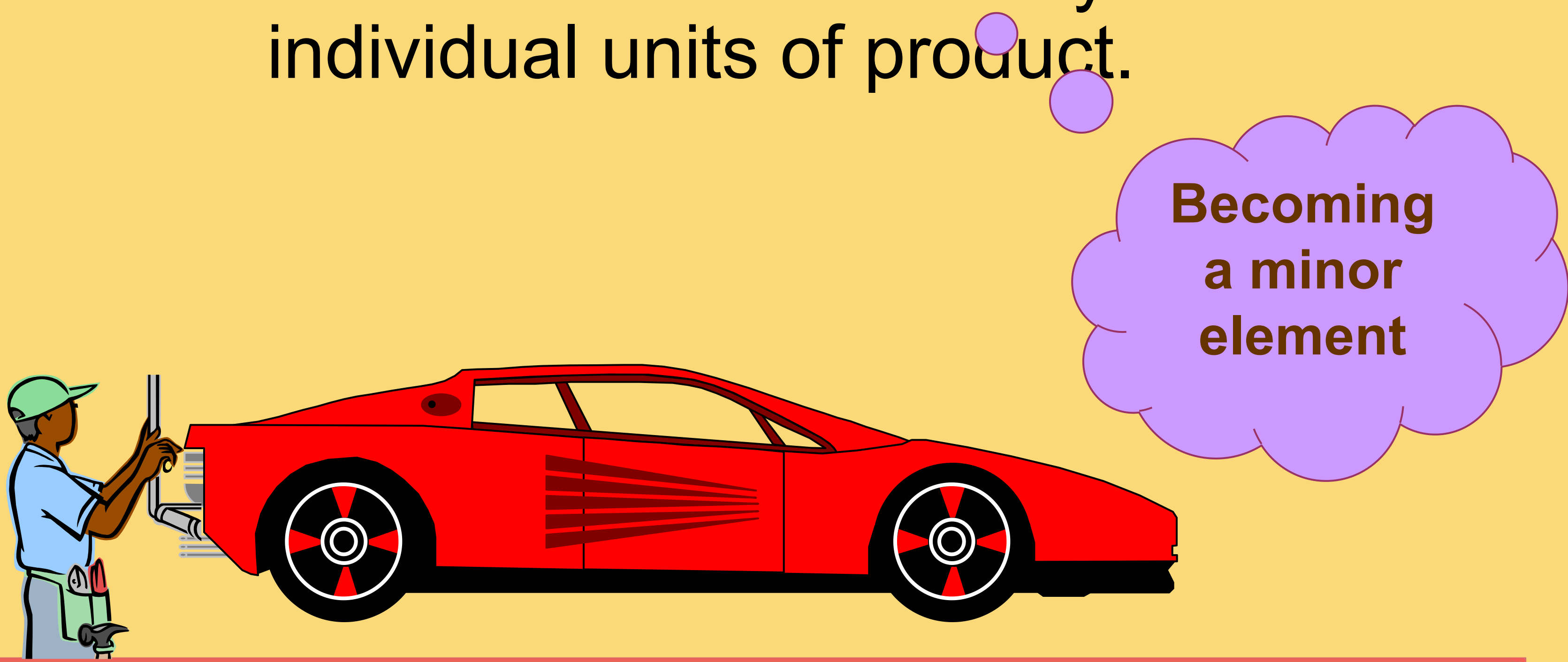
Raw materials that become an integral part of the product and that can be *conveniently* *and* *directly* traced to it.



**Example: A radio installed in an automobile**

# Direct Labor

Those labor costs that can be easily traced to individual units of product.




**Example:** **Wages paid to automobile assembly workers**

# Manufacturing Overhead

**Manufacturing costs that cannot be traced conveniently and directly to specific units produced.**

**Indirect materials, indirect labor, and other manufacturing overhead**

**Materials used to support the production process.**

**Examples:** lubricants and cleaning supplies used in the automobile assembly plant.

**Wages paid to employees who are not directly involved in production work.**

**Examples:** maintenance workers, janitors and security guards.

# Non-manufacturing Costs

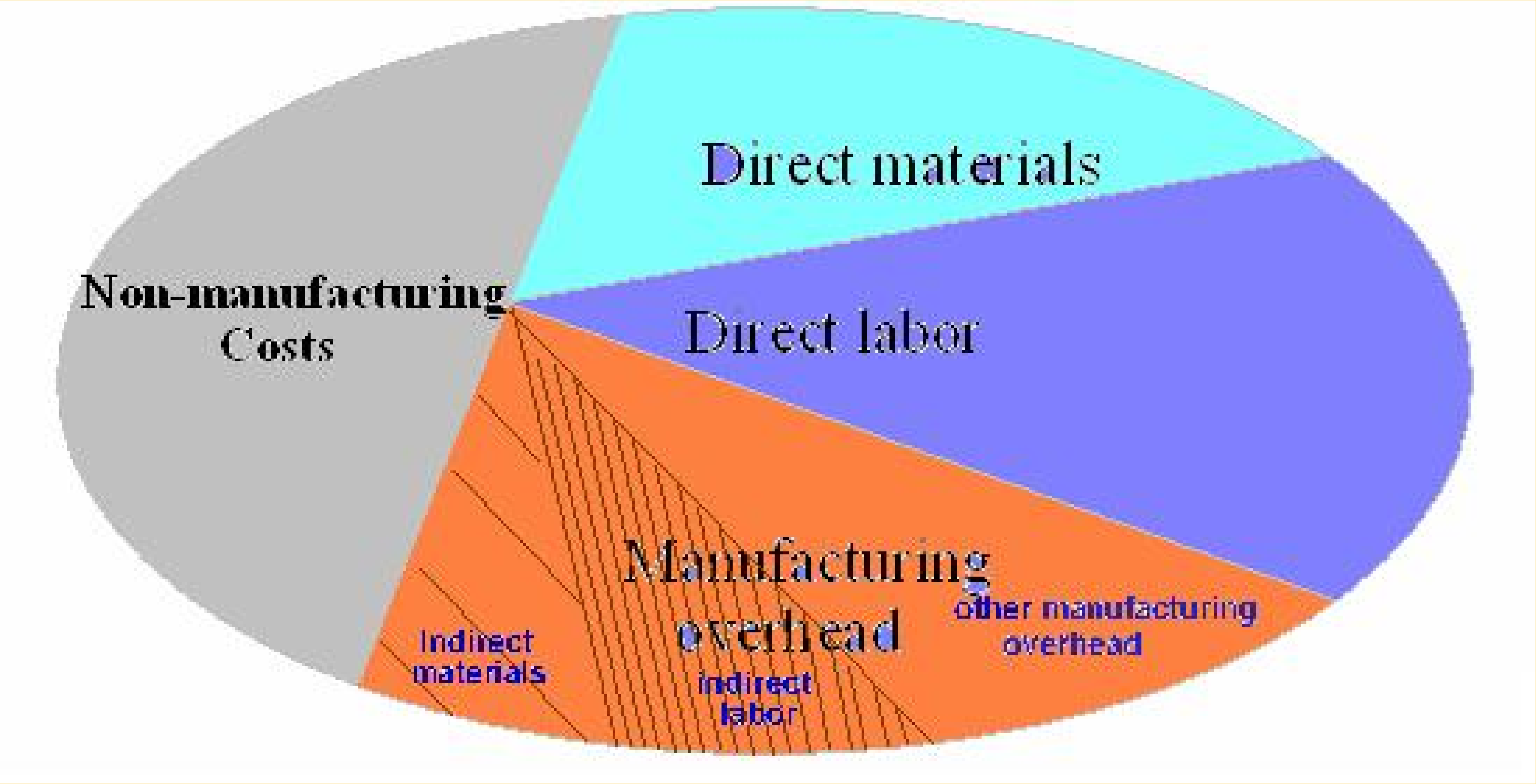
Selling Costs

Costs necessary to get the order and deliver the product.

Administrative Costs

All executive, organizational, and clerical costs.

# Cost Classifications



# Learning Objective 2

Distinguish between product costs and period costs and give examples of each.

# Matching Principle-----based on Accrual accounting

Costs incurred to generate a particular revenue should be recognized as expenses in the same period that the revenue is recognized.

Economic events are recognized by matching revenues to expenses (the matching principle) at the time in which the transaction occurs rather than when payment is made (or received).

# Product Costs Versus Period Costs

**Product costs include direct materials, direct labor, and manufacturing overhead.**

Inventory

Sale

Cost of Good Sold

Balance Sheet

Income Statement

**Period costs include all selling costs and administrative costs.**

Expense

Income Statement

# Quick Check ✓

Which of the following costs would be considered a period rather than a product cost in a manufacturing company?

A. Manufacturing equipment depreciation.

B. Property taxes on corporate headquarters.

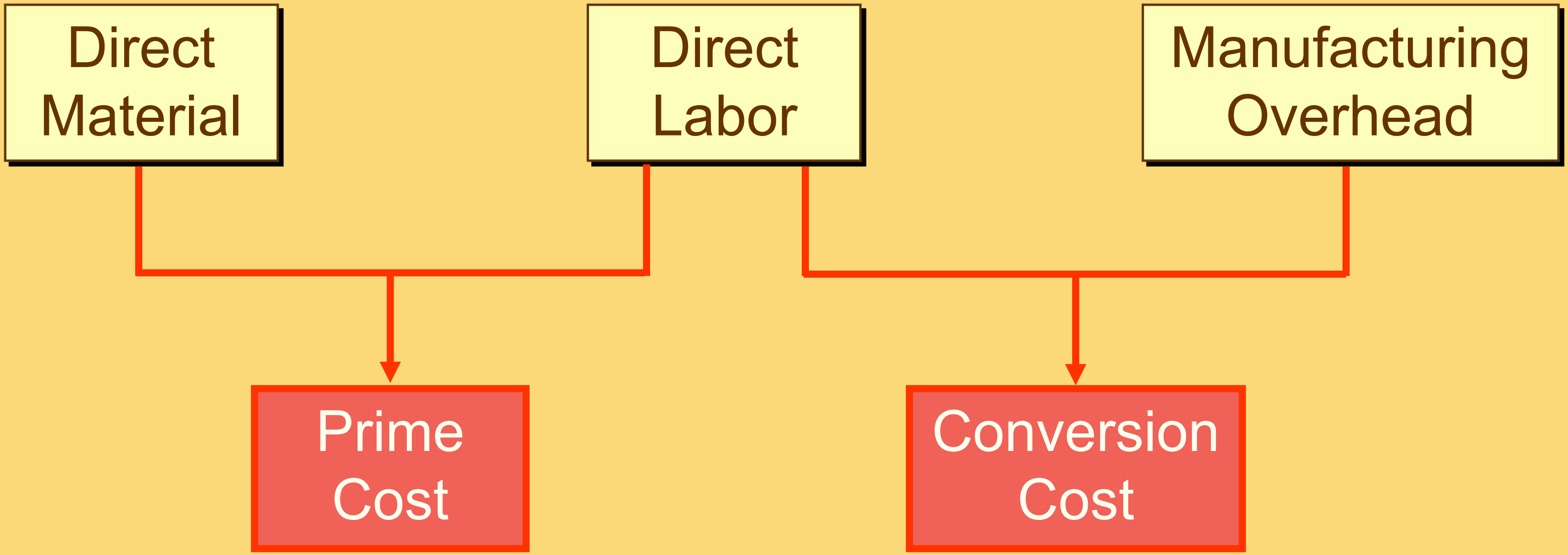
C. Direct materials costs.

D. Electrical costs to light the production facility.

E. Sales commissions.

# Classifications of Costs

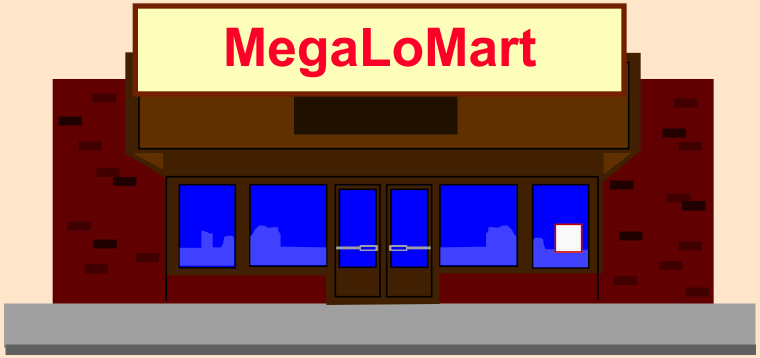
Manufacturing costs are often classified as follows:



# Comparing Merchandising and Manufacturing Activities

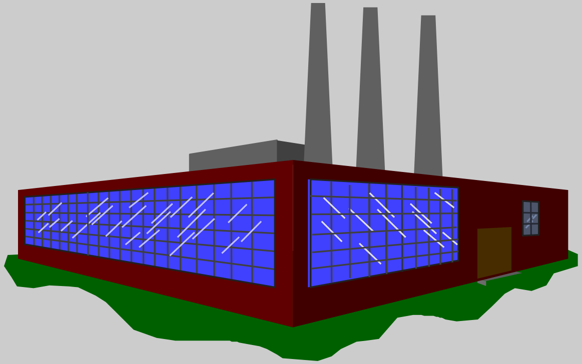
## Merchandisers . . .

- Buy finished goods.
- Sell finished goods.




## Manufacturers . . .

- Buy raw materials.
- Produce and sell finished goods.



# Balance Sheet

## Merchandiser

**Current assets**

- Cash
- Receivables
- Prepaid Expenses
- Merchandise Inventory

## Manufacturer

**Current Assets**

- Cash
- Receivables
- Prepaid Expenses
- Inventories
  - Raw Materials
  - Work in Process
  - Finished Goods

# Balance Sheet

## Merchandiser

**Current assets**

- Cash
- Receivables
- Prepaid Expenses

Partially complete products – some material, labor, or overhead has been added.

## Manufacturer

**Current Assets**

- Cash

Materials waiting to be processed.

- Inventories
  - Raw Materials
  - Work in Process
  - Finished Goods

Completed products awaiting sale.

# Learning Objective 3

Prepare an income statement including calculation of the cost of goods sold.

# The Income Statement

Cost of goods sold for manufacturers differs only slightly from cost of goods sold for merchandisers.

## Merchandising Company

| | |
|---|---|
| Cost of goods sold: | |
| Beg. merchandise inventory | $ 14,200 |
| + Purchases | 234,150 |
| Goods available for sale | $ 248,350 |
| - Ending merchandise inventory | (12,100) |
| = Cost of goods sold | $ 236,250 |

## Manufacturing Company

| | |
|---|---|
| Cost of goods sold: | |
| Beg. finished goods inv. | $ 14,200 |
| + Cost of goods manufactured | 234,150 |
| Goods available for sale | $248,350 |
| - Ending finished goods inventory | (12,100) |
| = Cost of goods sold | $236,250 |

# Basic Equation for Inventory Accounts

**Beginning balance** + **Additions to inventory** = **Ending balance** + **Withdrawals from inventory**

# Quick Check ✓

If your inventory balance at the beginning of the month was $1,000, you bought $100 during the month, and sold $300 during the month, what would be the balance at the end of the month?

A. $1,000.

B. $   800.

C. $1,200.

D. $   200.

# Learning Objective 4

Prepare a schedule of cost of goods manufactured.

# Schedule of Cost of Goods Manufactured

Calculates the cost of raw material, direct labor and manufacturing overhead used in production.

Calculates the manufacturing costs associated with goods that were finished during the period.

# Product Cost Flows

| Raw Materials | Manufacturing Costs | Work In Process |
|---|---|---|
| Beginning raw materials inventory | Direct materials | |
| + Raw materials purchased | | |
| = Raw materials available for use in production | | |
| – Ending raw materials inventory | | |
| = Raw materials used in production | | |

As items are removed from raw materials inventory and placed into the production process, they are called direct materials.

# Product Cost Flows

| Raw Materials | Manufacturing Costs | Work In Process |
|---|---|---|
| Beginning raw materials inventory | Direct materials | |
| + Raw materials purchased | + Direct labor | |
| = Raw materials available for use in production | + Mfg. overhead | |
| – Ending raw materials inventory | = Total manufacturing costs | |
| = Raw materials used in production | | |

Conversion costs are costs incurred to convert the direct material into a finished product.

# Product Cost Flows

**Raw Materials**

Beginning raw materials inventory
+ Raw materials purchased
= Raw materials available for use in production
– Ending raw materials inventory
= Raw materials used in production

**Manufacturing Costs**

Direct materials
+ Direct labor
+ Mfg. overhead
= Total manufacturing costs

**Work In Process**

Beginning work in process inventory
+ Total manufacturing costs
= Total work in process for the period

**All manufacturing costs incurred during the period are added to the beginning balance of work in process.**

# Product Cost Flows

## Raw Materials

Beginning raw materials inventory
\+ Raw materials purchased
= Raw materials available for use in production
– Ending raw materials

## Manufacturing Costs

Direct materials
\+ Direct labor
\+ Mfg. overhead
= Total manufacturing costs

## Work In Process

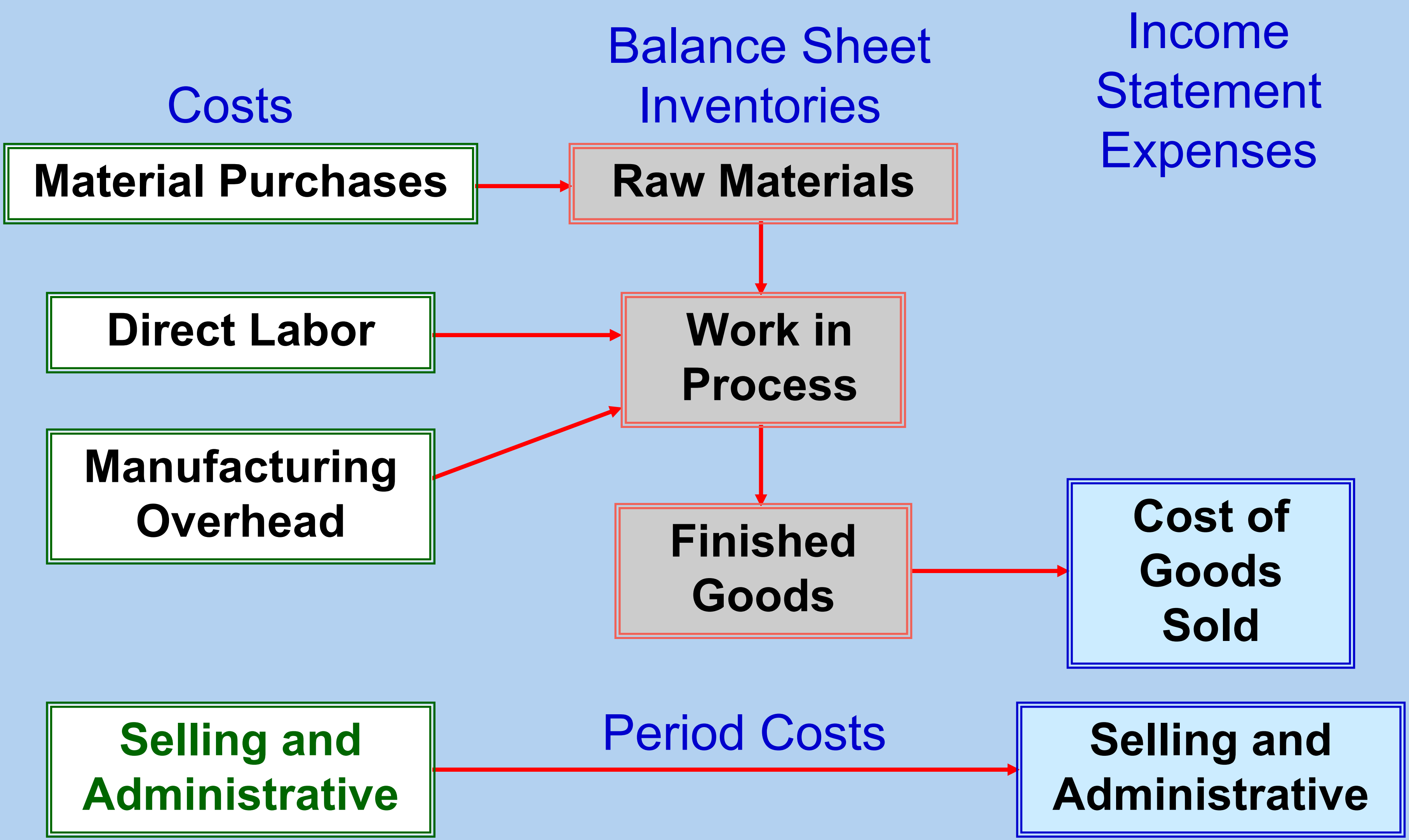
Beginning work in process inventory
\+ Total manufacturing costs
= Total work in process for the period
– Ending work in process inventory
= Cost of goods manufactured

**Costs associated with the goods that are completed during the period are transferred to finished goods inventory.**

# Product Cost Flows

| Work In Process | | Finished Goods |
|---|---|---|
| Beginning work in process inventory | | Beginning finished goods inventory |
| + Manufacturing costs for the period | | + Cost of goods manufactured |
| = Total work in process for the period | | = Cost of goods available for sale |
| – Ending work in process inventory | | - Ending finished goods inventory |
| = Cost of goods manufactured | | Cost of goods sold |

# Manufacturing Cost Flows



## Payout

Pay $100 for Raw Material

## Cost

The $100 listed on the balance sheet as an inventory

## Expense

The $100 expensed until the finished goods are sold that are made of the raw material

# Quick Check ✓

Beginning raw materials inventory was $32,000. During the month, $276,000 of raw material was purchased. A count at the end of the month revealed that $28,000 of raw material was still present. What is the cost of direct material used?

A. $276,000

B. $272,000

C. $280,000

D. $ 2,000

# Quick Check ✓

Direct materials used in production totaled $280,000. Direct labor was $375,000 and factory overhead was $180,000. What were total manufacturing costs incurred for the month?

A. $555,000

B. $835,000

C. $655,000

D. Cannot be determined.

# Quick Check ✓

Beginning work in process was $125,000. Manufacturing costs incurred for the month were $835,000. There were $200,000 of partially finished goods remaining in work in process inventory at the end of the month. What was the cost of goods manufactured during the month?

A. $1,160,000
B. $ 910,000
C. $ 760,000
D. Cannot be determined.

# Quick Check ✓

Beginning finished goods inventory was $130,000. The cost of goods manufactured for the month was $760,000. And the ending finished goods inventory was $150,000. What was the cost of goods sold for the month?

A. $ 20,000.

B. $740,000.

C. $780,000.

D. $760,000.

# Learning Objective 5

Understand the differences between variable costs and fixed costs.

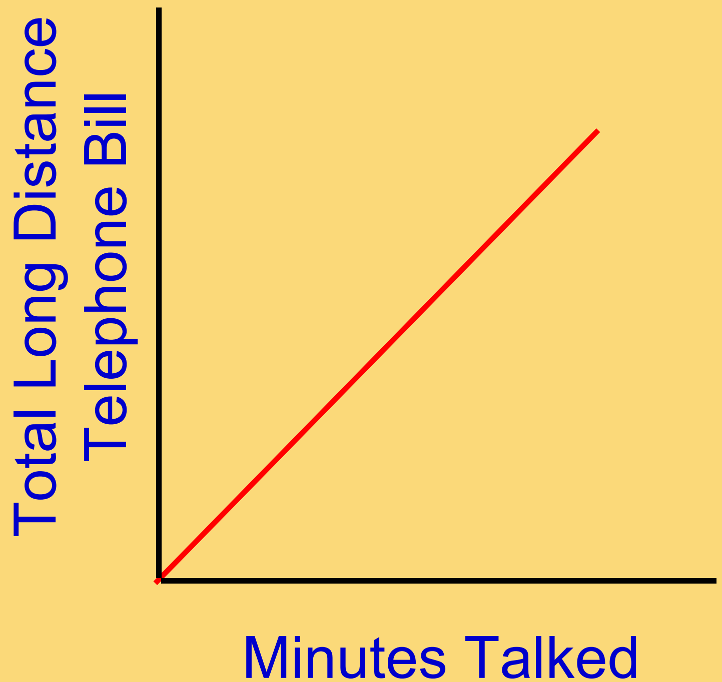
# Cost Classifications for Predicting Cost Behavior

How a cost will react to changes in the level of activity within the relevant range.

- Total **variable costs** change when activity changes.
- Total **fixed costs** remain unchanged when activity changes.

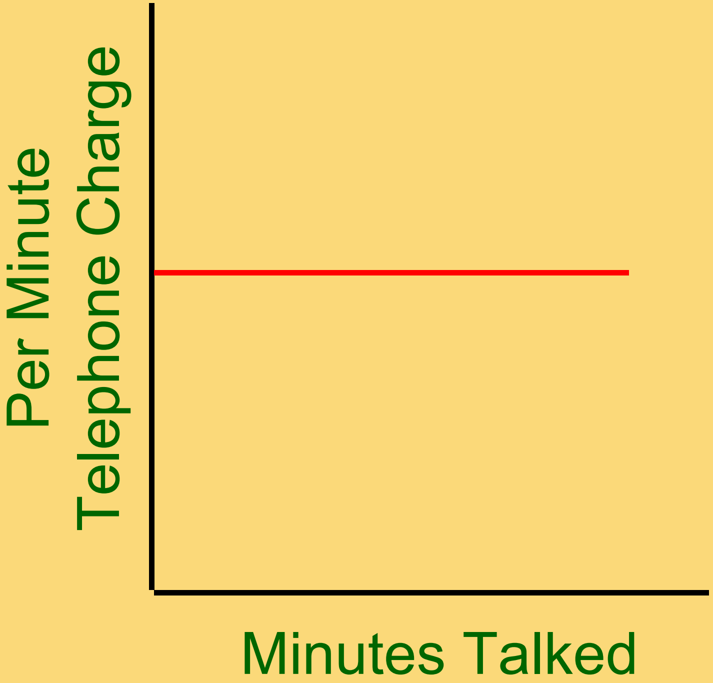
# Variable Cost

Your **total long distance** telephone bill is based on how many minutes you talk.



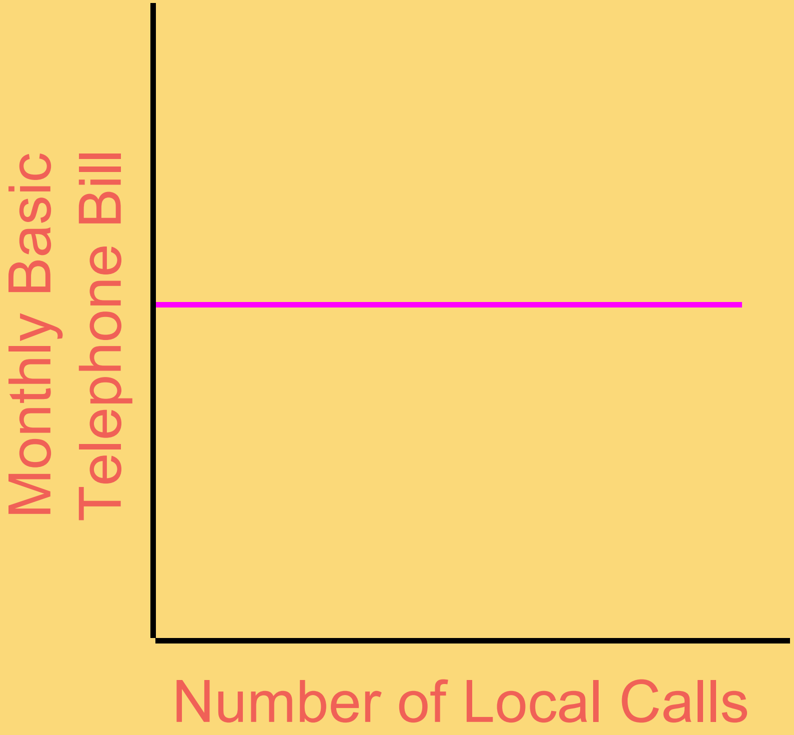

# Variable Cost Per Unit

The **cost per long distance minute** talked is constant.



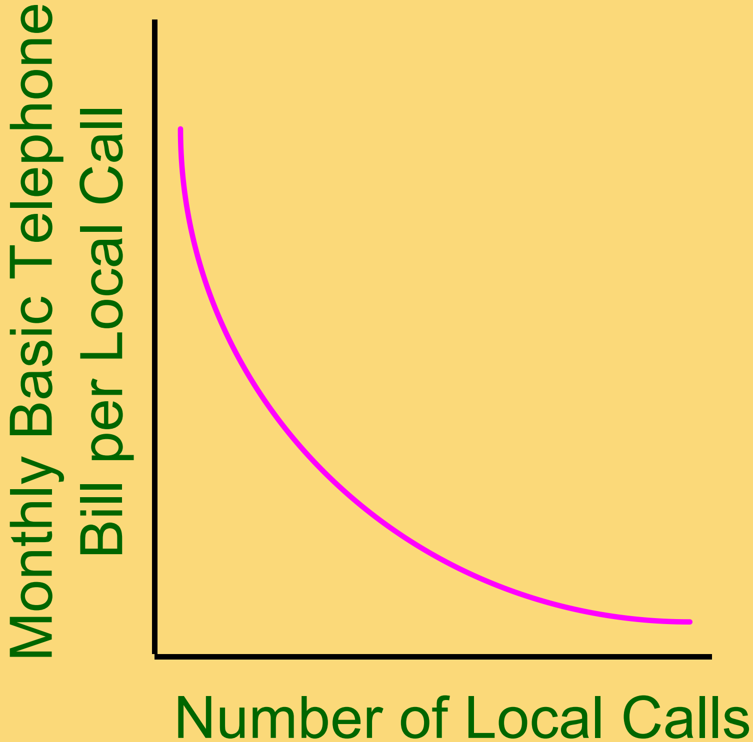
# Fixed Cost

Your monthly **basic telephone bill** probably does not change when you make more local calls.




# Fixed Cost Per Unit

The average fixed cost **per local call** decreases as more local calls are made.



# Cost Classifications for Predicting Cost Behavior

| Behavior of Cost (within the relevant range) | | |
|---|---|---|
| **Cost** | **In Total** | **Per Unit** |
| Variable | Total variable cost changes as activity level changes. | Variable cost per unit remains the same over wide ranges of activity. |
| Fixed | Total fixed cost remains the same even when the activity level changes. | Average fixed cost per unit goes down as activity level goes up. |

- Importance of understanding cost behavior
  - Food costs at a hotel
  - A hammer costs $1,000

# Quick Check ✓

Which of the following costs would be variable with respect to the number of cones sold at a Baskins & Robbins shop? (There may be more than one correct answer.)

A. The cost of lighting the store.

B. The wages of the store manager.

C. The cost of ice cream.

D. The cost of napkins for customers.

# Learning Objective 6

Understand the differences between direct and indirect costs.

# Cost Object

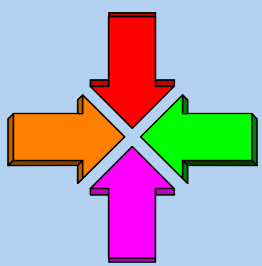
Anything for which cost data are desired.

Examples: products, customers, jobs, organizational subunits, etc.

# Assigning Costs to Cost Objects

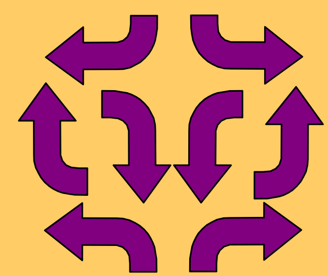
## Direct costs

- Costs that can be easily and conveniently traced to a unit of product or other cost object.
- Examples: direct material and direct labor



## Indirect costs

- Costs that cannot be easily and conveniently traced to a unit of product or other cost object.
- Example: manufacturing overhead



# Your Judgment

- ***Common cost***
  - A cost that is incurred to support a number of cost objects but cannot be traced to them individually.

- ***An Example-----A trade with DOD***

# Learning Objective 7

Define and give examples of cost classifications used in making decisions: differential costs, opportunity costs, and sunk costs.

# Cost Classifications for Decision Making

- Every decision involves a choice between at least two alternatives.
- Only those costs and benefits that differ between alternatives are relevant in a decision. All other costs and benefits can and should be ignored.

# Differential Cost and Revenue

## Costs and revenues that differ among alternatives.

**Example:** You have a job paying $1,500 per month in your hometown. You have a job offer in a neighboring city that pays $2,000 per month. The commuting cost to the city is $300 per month.

**Differential revenue is:**
**$2,000 – $1,500 = $500**

**Differential cost is:**
**$300**

# Opportunity Cost

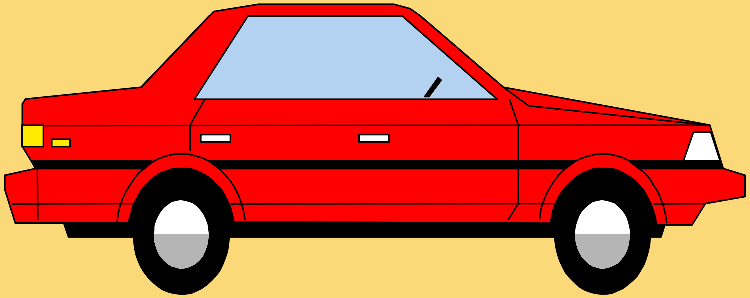
**The potential benefit that is given up when one alternative is selected over another.**

Example: If you were not attending college, you could be earning RMB100,000 per year. Your opportunity cost of attending college for one year is RMB100,000.

# Sunk Costs

**Sunk costs have already been incurred and cannot be changed now or in the future. They should be ignored when making decisions.**

Example: You bought an automobile that cost $10,000 two years ago. The $10,000 cost is sunk because whether you drive it, park it, trade it, or sell it, you cannot change the $10,000 cost.



# Quick Check ✓

Suppose you are trying to decide whether to drive or take the train to Portland to attend a concert. You have ample cash to do either, but you don't want to waste money needlessly. Is the cost of the train ticket relevant in this decision? In other words, should the cost of the train ticket affect the decision of whether you drive or take the train to Portland?

A. Yes, the cost of the train ticket is relevant.

B. No, the cost of the train ticket is not relevant.

# Quick Check ✓

Suppose you are trying to decide whether to drive or take the train to Portland to attend a concert. You have ample cash to do either, but you don't want to waste money needlessly. Is the annual cost of licensing your car relevant in this decision?

A. Yes, the licensing cost is relevant.

B. No, the licensing cost is not relevant.

# Quick Check ✓

Suppose that your car could be sold now for $5,000. Is this a sunk cost?

A. Yes, it is a sunk cost.

B. No, it is not a sunk cost.

# Summary of the Types of Cost Classifications

- Financial reporting
- Predicting cost behavior
- Assigning costs to cost objects
- Decision making

# End of Chapter 2